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Chapter 1

Introduction to Lab

1.1 Tools

These are different tools that will be used to complete the lab exercises.

- MySQL on QMUL's Unified Platform
- Oracle Live Sql

Oracle Live Sql is the simply solution and doesn't require any installation, simply you need to upload the labtables.sqlscript. That is shown in the next chapter. But it has limitation and you will not be able to fully explore the oracle database

The next chapters will get you set up for your labs on MySQL and oracle live sql. We strongly encourage you to do all exercises on both systems in order to become acquainted with different DBMSs.

1.2 Outline of the Manual

Chapter 2: MySQL on QMUL's Unified Platform

This will guide you how to setup MySQL on QMUL's Unified Platform. This is your first lab. You need to follow all the steps to build MySQL on the Unified Platform and run your labs

Chapter 3: Oracle Live SQL Environment

This will guide you how to use the oracle live sql environment and upload the labtable.sql script file.

Chapter 4: Installing Oracle Database XE on Windows 10.

This will guide you how to locally install oracle database XE on Windows 10 machine. This is highly recommended to explore the full potential of an oracle database.

Chapter 5: Oracle Database on Mac OS using Docker

This will guide you how to locally install oracle database on Mac OS. As Oracle database is not available for Mac platform, so we try to use the Docker container.

Chapter 6: Lab Exercises.

All lab exercises are given starting from Week-2.

Appendixes: labtables.sql.

It is recommended to download the labtables.sql from qmplus or save appendix contents to a script file

Chapter 2

MySQL on QMUL's Unified Platform

2.1 Introduction to the Unified Platform

QMUL's Unified Platform is designed for developing and running containerized applications – MySQL instances in the case of this course. It relies on Kubernetes, also known as K8s, an open-source container orchestration system for automating software deployment, scaling, and management. In Kubernetes, containers run inside units called pods. In what follows you will be shown how to create a pod in which to run an instance of MySQL.

2.2 MySQL on the Unified Platform

QMUL's Unified Platform can be accessed from the link

`https://unified.a.comp-teach.qmul.ac.uk/docs`

Step-1: A sign-in pop-up window should appear, otherwise click on 'Sign in'.

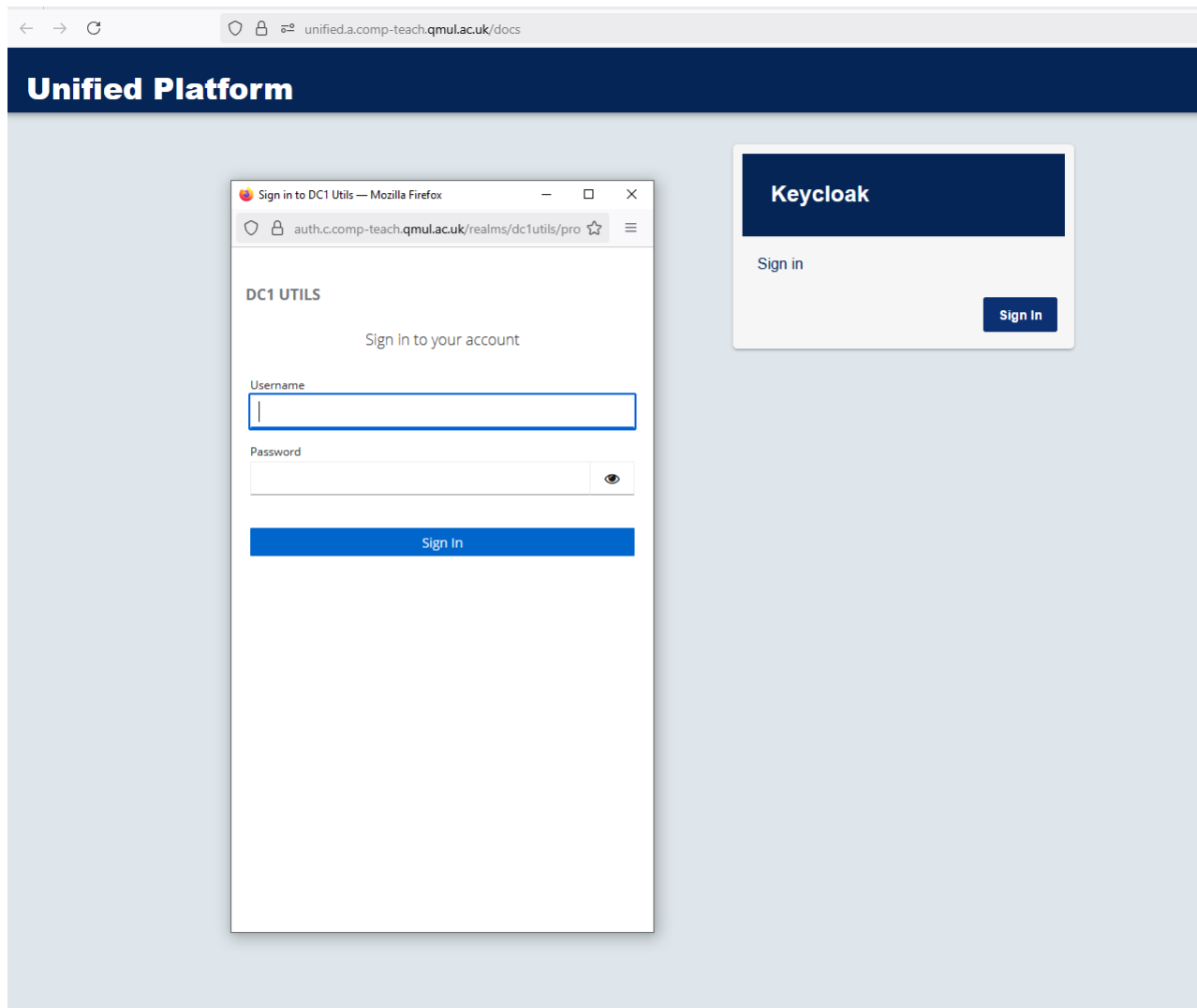


Figure 2.1: Unified Platform Login Page

Step-2: Insert your QMUL credentials and click on ‘Sign in’.

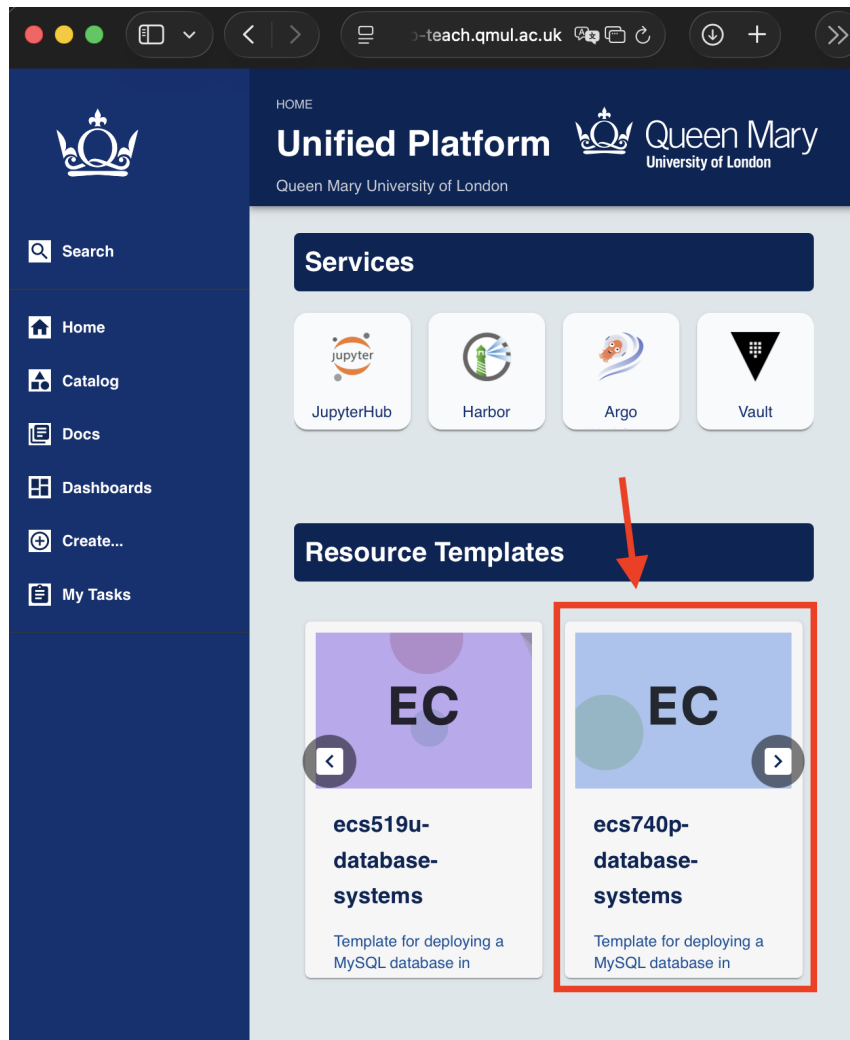


Figure 2.2: Unified Platform Home Page

Step-3: Click **Home** from left menu

Step-4: Click on **ecs740p-database-systems**

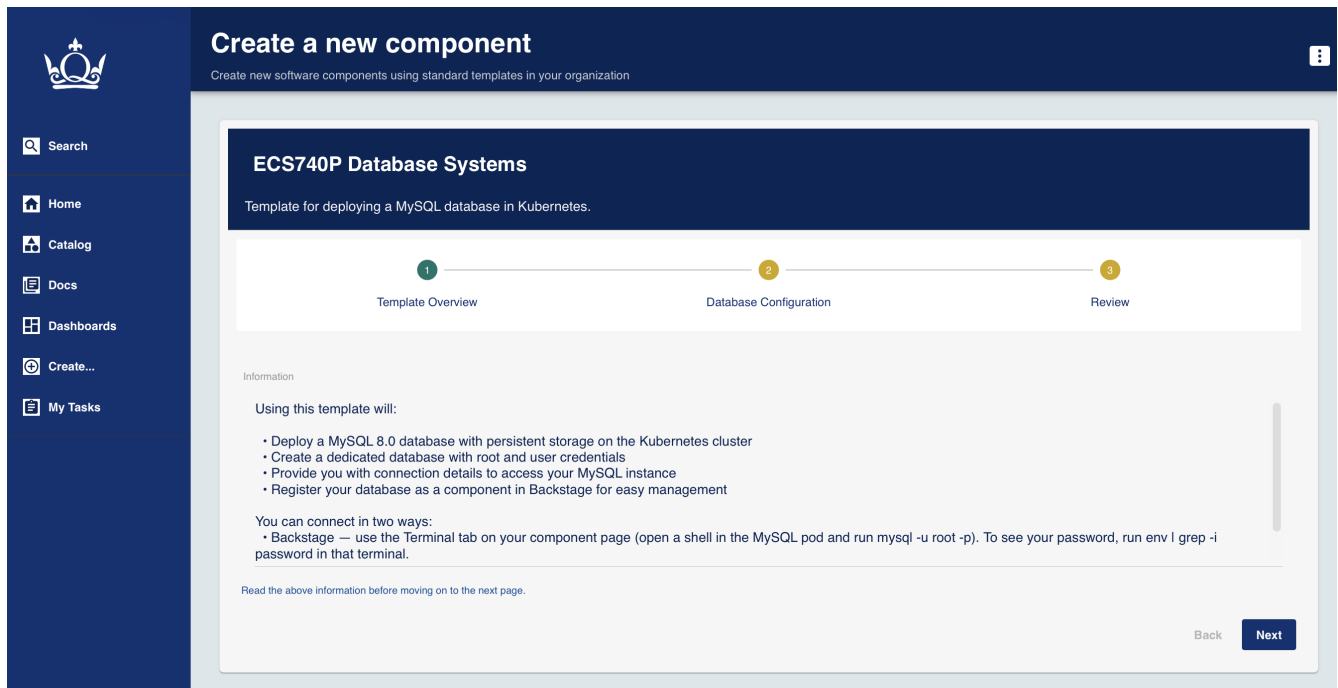


Figure 2.3: Instructions

Step-5: Read the instructions! In fact it's a good idea to copy and past them somewhere for future reference.

ECS740P Database Systems

Template for deploying a MySQL database in Kubernetes.

1 Template Overview 2 Database Configuration

Deployment Intent *

New deployment

Choose whether this run creates a new deployment or deletes resources

Instance Name *

mydb

Name for your MySQL deployment. Used for Kubernetes resource naming (e.g. mysql-(name)-(username)). If you only need one database, the default is fine.

Database Name

ecs740p

The MySQL database name to create. This is what you USE when connected.

Database Username

Username for the database user account. Leave blank to auto-generate.

Database Password

Password for the database user account. Leave blank to auto-generate.

Database Root Password

Root password for the MySQL instance. Leave blank to auto-generate.

Figure 2.4: Database configuration

Step-6: Give your instance a name, choose a user name, a user password and a root password. Leave the Database Name as **ecs740p**: this is the database that the lab script creates and uses. You can leave everything as it is if you prefer, in which case the system will choose names and passwords automatically for you. All configuration details will be available to you (see below) whether you choose them yourself or let the system do it for you.

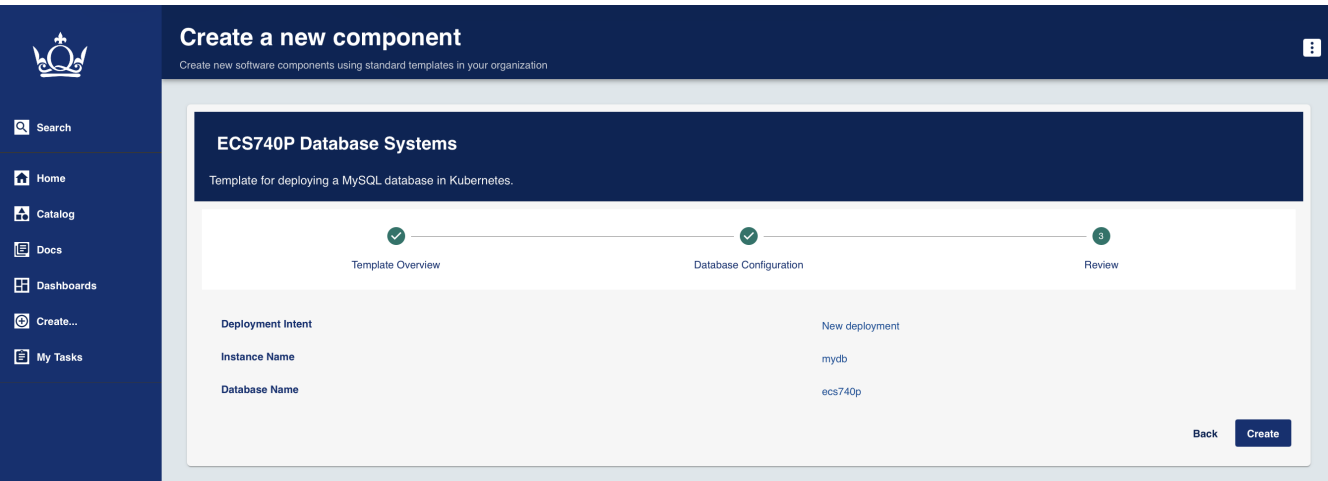


Figure 2.5: Review and create

Step-7: Review and click on **Create**

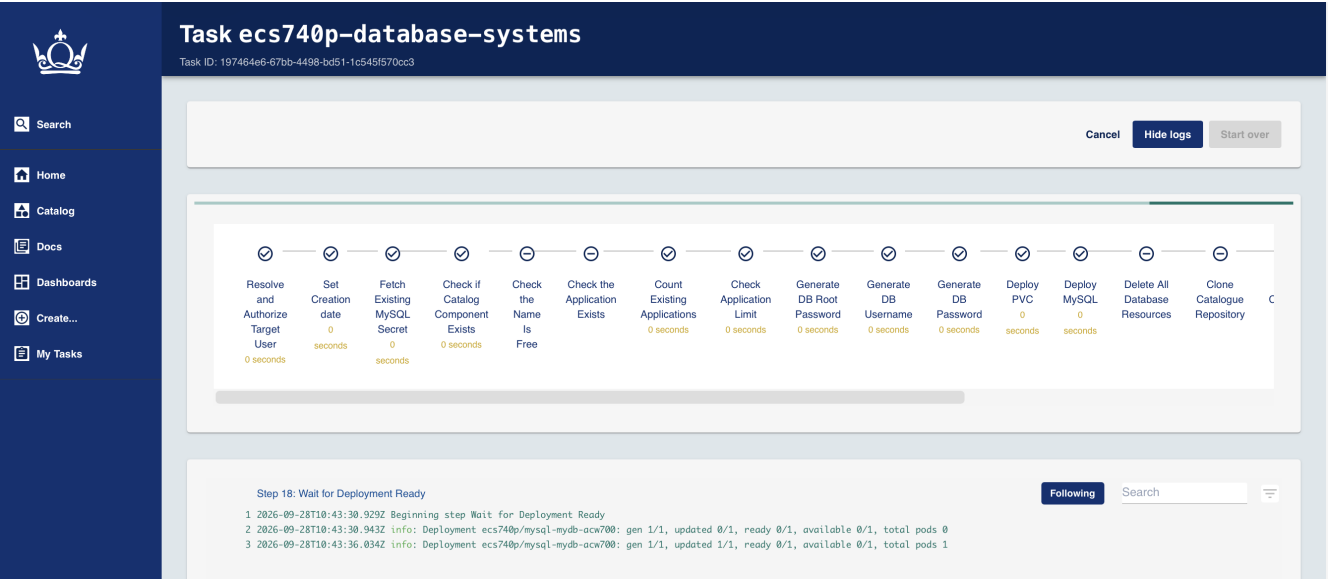


Figure 2.6: Instance creation

Step-8: The system will create your instance. This may take a while.

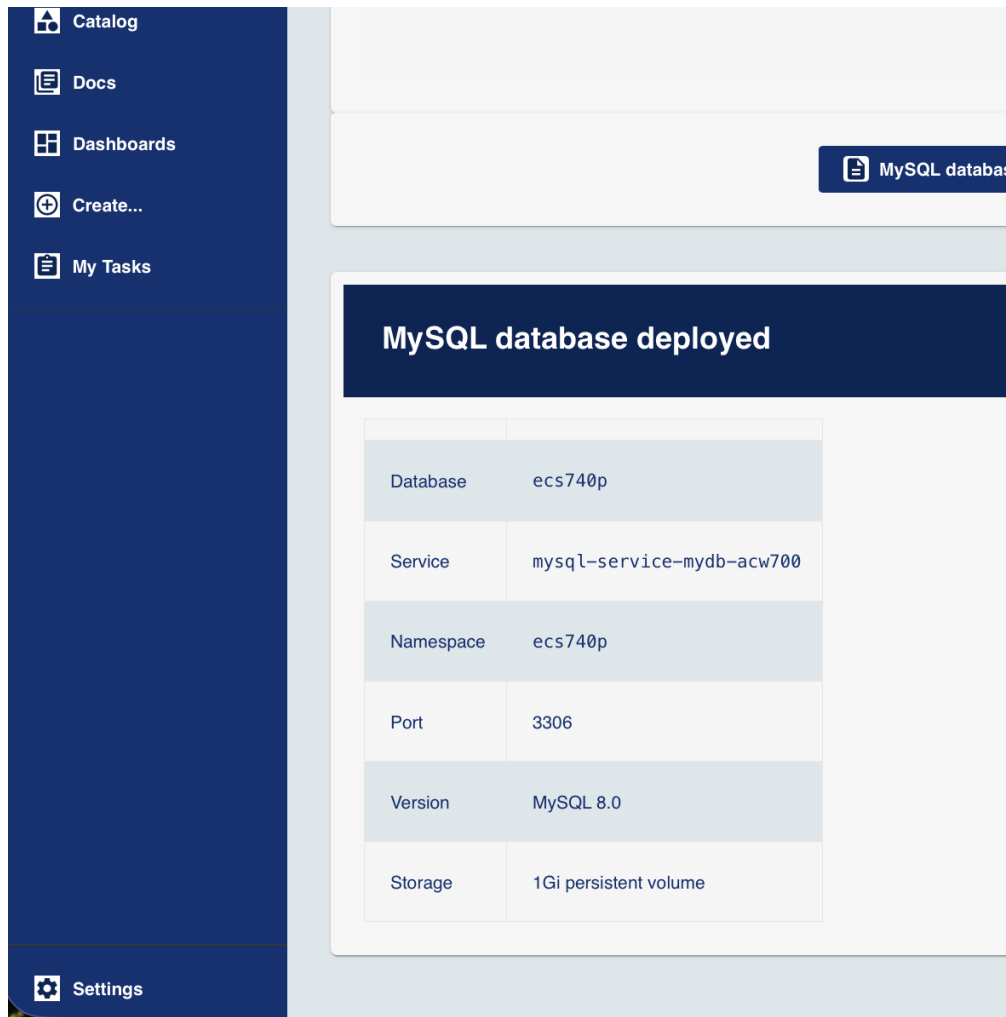


Figure 2.7: Instance successfully created

Step-9: When the system is finished, you should see something that looks like Figure 2.7 if you scroll down to the bottom of the page



Search

Home

Catalog

Docs

Dashboards

Create...

My Tasks

Queen Mary Catalog

Create

Support

Kind

Component

Type

all

PERSONAL

Owned 7

Starred 0

QUEEN MARY

All 7

OWNER

LIFECYCLE

Owned components (7)

Filter

Name	System	Owner	Type	Lifecycle	Description	Tags	Actions
GPU capacity vs free	EECS Research	eecs-research	dashboard	production	Real-time GP...	eecs-research, gpu, observability, dashboard	🔗 ✎ ☆
GPU Usage: Networks		networks	dashboard	production	GPU utilisatio...	eecs-research, gpu, observability, dashboard	🔗 ✎ ☆
GPU Usage: School		school	dashboard	production	GPU utilisatio...	eecs-research, gpu, observability, dashboard	🔗 ✎ ☆
How-to-write-Docs-on-this-site		eecs-research	service	experimental	Documentatio...		🔗 ✎ ☆
kubeflow-Docs		eecs-research	website	experimental	Kubeflow...		🔗 ✎ ☆
mysql-db-mysql-acw700	ecs519u	Andres Aharhel Mercado Velazquez	service	experimental	A MySQL...	database-systems, mysql	🔗 ✎ ☆
pg-mysql-db-mysql-acw700	ecs740p	Andres Aharhel Mercado Velazquez	service	experimental	A MySQL...	database-systems, mysql, ecs740p	🔗 ✎ ☆

Figure 2.8: Catalog

Step-10: Click on **Catalog** in the left-hand side navigation bar. You should now see the component that you have just created (last line in Figure 2.8).

Step-11: Click on the name of your component.

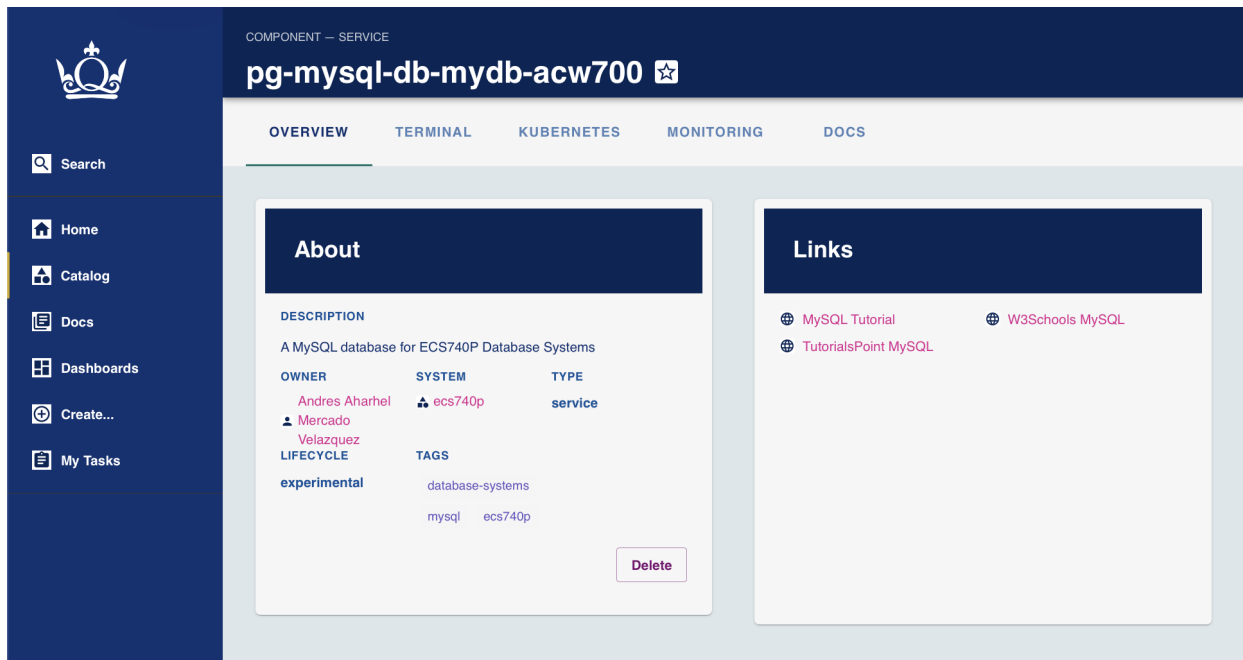


Figure 2.9: Your component

Step-12: You'll see the details of your component as well as some useful links.

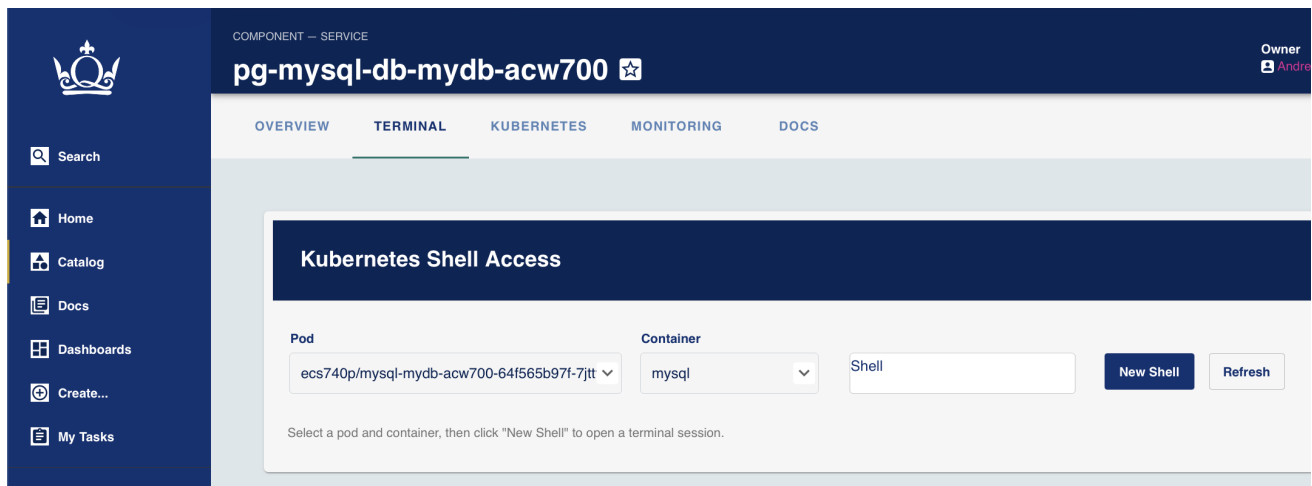


Figure 2.10: Create a new shell

Step-13: Click on **TERMINAL** in the top navigation bar, then click on **New Shell**.

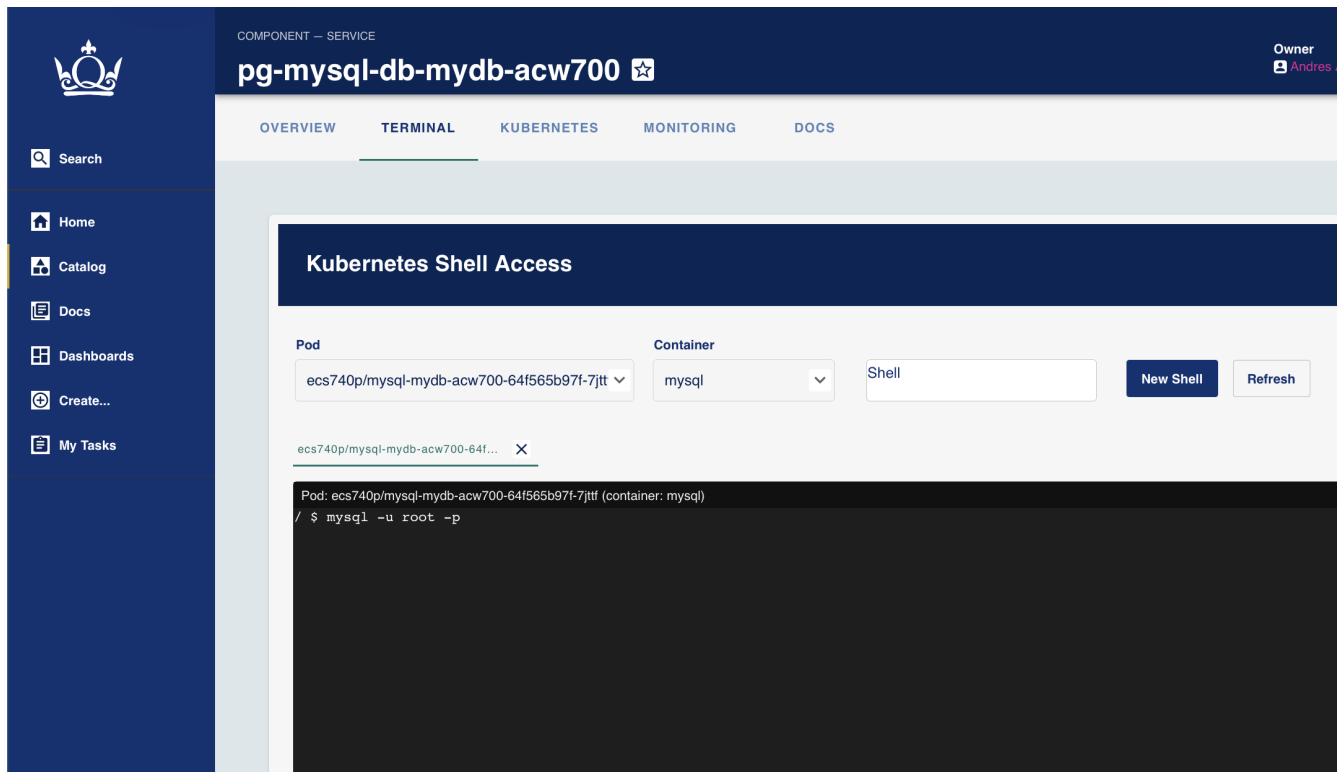


Figure 2.11: MySQL Command Prompt

Step-14: Type `mysql -u root -p` and then enter your *root* password.

If you don't know your root password – e.g. you've left the system choose one for you – then start by running the following command `env | grep -i password`. This will display your local user password and your root user password. If you don't know the name of your local user run the command `env | grep -i user` (root is not displayed as it's just called *root*).

Note: the system has automatically created two users for you: a local user which represents a user of the database, and a root user which represents an administrator of the database. The local user can read from the database, but is not allowed to create tables, delete tables, etc. In order to create the database and its tables you need to choose the root user. You will learn about the different users, and how to set their privileges in this course.

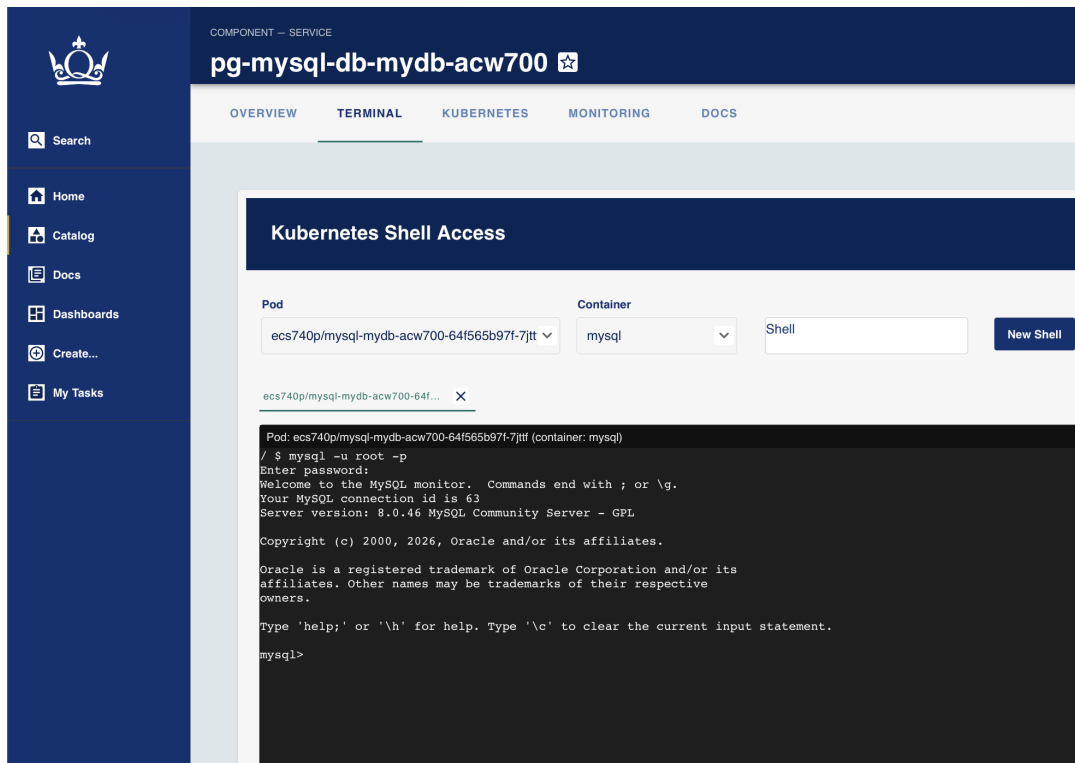


Figure 2.12: MySQL is running

Step-15: MySQL is now running in the terminal and you will be able to create the database used for the labs. Note: you can run several shells side-by-side, e.g. one as root and one as a local user.

2.3 Creating the database

2.3.1 Get the MySQL script on QMPlus

Download the [labtables.sql \(for MySQL\)](https://qmplus.qmul.ac.uk/course/view.php?id=33349) script file from the course's QMPlus page <https://qmplus.qmul.ac.uk/course/view.php?id=33349>. Open the file in a text editor. This is what it contains:

Listing 2.1: labtables.sql (for MySQL)

```
1 CREATE DATABASE IF NOT EXISTS 'ecs740p';
2 USE 'ecs740p';
3 select 'Building demonstration tables. Please wait.' as '';
4
5 DROP TABLE IF EXISTS EMP;
6 DROP TABLE IF EXISTS DEPT;
7 DROP TABLE IF EXISTS BONUS;
8 DROP TABLE IF EXISTS SALGRADE;
9 DROP TABLE IF EXISTS DUMMY;
```

```

10
11
12 CREATE TABLE EMP
13 (EMPNO SMALLINT NOT NULL,
14  ENAME VARCHAR(10),
15  JOB VARCHAR(9),
16  MGR SMALLINT,
17  HIREDATE DATETIME,
18  SAL DECIMAL(7, 2),
19  COMM DECIMAL(7, 2),
20  DEPTNO TINYINT);
21
22 INSERT INTO EMP VALUES
23 (7369, 'SMITH', 'CLERK', 7902,
24  STR_TO_DATE('17-DEC-1980', '%d-%M-%Y'), 800, NULL, 20);
25 INSERT INTO EMP VALUES
26 (7499, 'ALLEN', 'SALESMAN', 7698,
27  STR_TO_DATE('20-FEB-1981', '%d-%M-%Y'), 1600, 300, 30);
28 INSERT INTO EMP VALUES
29 (7521, 'WARD', 'SALESMAN', 7698,
30  STR_TO_DATE('22-FEB-1981', '%d-%M-%Y'), 1250, 500, 30);
31 INSERT INTO EMP VALUES
32 (7566, 'JONES', 'MANAGER', 7839,
33  STR_TO_DATE('2-APR-1981', '%d-%M-%Y'), 2975, NULL, 20);
34 INSERT INTO EMP VALUES
35 (7654, 'MARTIN', 'SALESMAN', 7698,
36  STR_TO_DATE('28-SEP-1981', '%d-%M-%Y'), 1250, 1400, 30);
37 INSERT INTO EMP VALUES
38 (7698, 'BLAKE', 'MANAGER', 7839,
39  STR_TO_DATE('1-MAY-1981', '%d-%M-%Y'), 2850, NULL, 30);
40 INSERT INTO EMP VALUES
41 (7782, 'CLARK', 'MANAGER', 7839,
42  STR_TO_DATE('9-JUN-1981', '%d-%M-%Y'), 2450, NULL, 10);
43 INSERT INTO EMP VALUES
44 (7788, 'SCOTT', 'ANALYST', 7566,
45  STR_TO_DATE('09-DEC-1982', '%d-%M-%Y'), 3000, NULL, 20);
46 INSERT INTO EMP VALUES
47 (7839, 'KING', 'PRESIDENT', NULL,
48  STR_TO_DATE('17-NOV-1981', '%d-%M-%Y'), 5000, NULL, 10);
49 INSERT INTO EMP VALUES
50 (7844, 'TURNER', 'SALESMAN', 7698,
51  STR_TO_DATE('8-SEP-1981', '%d-%M-%Y'), 1500, 0, 30);
52 INSERT INTO EMP VALUES
53 (7876, 'ADAMS', 'CLERK', 7788,
54  STR_TO_DATE('12-JAN-1983', '%d-%M-%Y'), 1100, NULL, 20);
55 INSERT INTO EMP VALUES
56 (7900, 'JAMES', 'CLERK', 7698,
57  STR_TO_DATE('3-DEC-1981', '%d-%M-%Y'), 950, NULL, 30);
58 INSERT INTO EMP VALUES
59 (7902, 'FORD', 'ANALYST', 7566,

```

```

60 STR_TO_DATE('3-DEC-1981', '%d-%M-%Y'), 3000, NULL, 20);
61 INSERT INTO EMP VALUES
62 (7934, 'MILLER', 'CLERK', 7782,
63 STR_TO_DATE('23-JAN-1982', '%d-%M-%Y'), 1300, NULL, 10);
64
65
66 CREATE TABLE DEPT
67 (DEPTNO TINYINT,
68 DNAME VARCHAR(14),
69 LOC VARCHAR(13) );
70
71 INSERT INTO DEPT VALUES (10, 'ACCOUNTING', 'NEW_YORK');
72 INSERT INTO DEPT VALUES (20, 'RESEARCH', 'DALLAS');
73 INSERT INTO DEPT VALUES (30, 'SALES', 'CHICAGO');
74 INSERT INTO DEPT VALUES (40, 'OPERATIONS', 'BOSTON');
75
76
77 CREATE TABLE BONUS
78 (ENAME VARCHAR(10),
79 JOB VARCHAR(9),
80 SAL DOUBLE,
81 COMM DOUBLE);
82
83
84 CREATE TABLE SALGRADE
85 (GRADE DOUBLE,
86 LOSAL DOUBLE,
87 HISAL DOUBLE);
88
89 INSERT INTO SALGRADE VALUES (1, 700, 1200);
90 INSERT INTO SALGRADE VALUES (2, 1201, 1400);
91 INSERT INTO SALGRADE VALUES (3, 1401, 2000);
92 INSERT INTO SALGRADE VALUES (4, 2001, 3000);
93 INSERT INTO SALGRADE VALUES (5, 3001, 9999);
94
95
96 CREATE TABLE DUMMY
97 (DUMMY DOUBLE);
98
99 INSERT INTO DUMMY VALUES (0);
100
101 COMMIT;
102
103 select 'PROMPT_Demonstration_table_build_is_complete.' as '';

```

2.3.2 Executing the MySQL script

The script will create a database schema `ecs740p` with five tables i.e. `EMP`, `DEPT`, `BONUS`, `SALGRADE` and `DUMMY`. Let's run it one step at a time to understand what it does. Copy and paste the lines 1 to 9 of Listing 2.1

in your terminal and press return:

```
Pod: ecs740p/mysql-mysdb-acw700-64f565b97f-7jttf (container: mysql)
mysql> USE `ecs740p`;
Database changed
mysql> select 'Building demonstration tables. Please wait.' as '';
+-----+
| Building demonstration tables. Please wait. |
+-----+
1 row in set (0.00 sec)

mysql>
mysql> DROP TABLE IF EXISTS EMP;
Query OK, 0 rows affected, 1 warning (0.01 sec)

mysql> DROP TABLE IF EXISTS DEPT;
Query OK, 0 rows affected, 1 warning (0.02 sec)

mysql> DROP TABLE IF EXISTS BONUS;
Query OK, 0 rows affected, 1 warning (0.01 sec)

mysql> DROP TABLE IF EXISTS SALGRADE;
Query OK, 0 rows affected, 1 warning (0.02 sec)

mysql> DROP TABLE IF EXISTS DUMMY;
Query OK, 0 rows affected, 1 warning (0.01 sec)

mysql>
```

Figure 2.13: Creating the database

The code is self-explanatory: you've created a database called `ecs740p` (unless it already existed), told SQL that you want to work in this database (that's what `USE 'ecs740p';` does), and made sure that no tables called `EMP`, `DEPT`, `BONUS`, `SALGRADE`, `DUMMY` already exist in the database (`DROP` will delete them if they do). The reason for these `DROP` commands is that we're going to create tables with these names, and if the names were already used the script would throw an error. You can see that the database `ecs740p` has been created successfully by executing the command `show databases;`.

```

root@ecs740p:mysql:/mysql-5.6.16-04363537-1.7.16 (container:mysql)
->          STR_TO_DATE('12-JAN-1983', '%d-%M-%Y'), 1100, NULL, 20);
Query OK, 1 row affected (0.03 sec)

mysql> INSERT INTO EMP VALUES
->          (7900, 'JAMES', 'CLERK', 7698,
->          STR_TO_DATE('13-DEC-1981', '%d-%M-%Y'), 950, NULL, 30);
Query OK, 1 row affected (0.03 sec)

mysql> INSERT INTO EMP VALUES
->          (7902, 'FORD', 'ANALYST', 7566,
->          STR_TO_DATE('13-DEC-1981', '%d-%M-%Y'), 3000, NULL, 20);
Query OK, 1 row affected (0.02 sec)

mysql> INSERT INTO EMP VALUES
->          (7934, 'MILLER', 'CLERK', 7782,
->          STR_TO_DATE('23-JAN-1982', '%d-%M-%Y'), 1300, NULL, 10);
Query OK, 1 row affected (0.02 sec)

mysql>
mysql> show tables;.
+-----+
| Tables_in_ecs740p |
+-----+
| EMP                |
+-----+
1 row in set (0.01 sec)

```

Figure 2.14: Creating a table

Next, let's create the first table. Copy and paste lines 12-63 of Listing 2.1 into the shell and hit return. You've created the table `EMP`, and you've then inserted rows of data into. You can check that it's been created correctly by executing the command `show tables;`, you can show the structure of the table by executing the command `desc EMP;`, and you can display the entire table by executing the command `select * from EMP;`.

Finally, copy and paste the rest of Listing 2.1 into the shell and hit return. This will create and populate the remaining tables. You are now ready to do the lab exercises.

When you're done simply execute the command `exit;` to leave MySQL.

2.4 Basic SQL Help

More information and troubleshooting help can be found by clicking on the Docs section of the left-hand-side navigation bar and then clicking on [ECS740P Database Systems](#).



Documentation

Documentation available in Queen Mary

- Home
- Catalog
- Docs
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- Create...
- My Tasks

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- Owned 6
- Starred 0

QUEEN MARY

- All 6

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TAGS

Owned (6)

Document	Owner
ECS519U Database Systems Template for deploying a MySQL database in Kubernetes.	ECS519U
ECS740P Database Systems Template for deploying a MySQL database in Kubernetes.	ECS740P
How-to-write-Docs-on-this-site Documentation on how to write Docs on this site.	eecs-research
kubeflow-Docs Kubeflow Documentation	eecs-research
mysql-db-mydb-acw700 A MySQL database for ECS519U Database Systems	Andres Aharhel Mercado Velazquez
pg-mysql-db-mydb-acw700 A MySQL database for ECS740P Database Systems	Andres Aharhel Mercado Velazquez

Figure 2.15: Navigating to the documentation for ECS740P.

Meanwhile, a good basic introduction to the language which you can use to assist with the lab exercises can be found here. You can also refer to the lecture notes for this week’s lecture.

Chapter 3

Oracle Live SQL Environment

3.1 Introduction

Oracle LiveSQL is a **cloud based service** that provides access to an Oracle database instance. No installation required. LiveSQL is especially useful for learning to program Oracle databases using Structured Query Language (**SQL**) and **PL/SQL**, Oracle's procedural language extension to SQL.

You simply need to upload database scripts labtables.sql and all your tables will be available and you can create your sql queries.

LiveSQL has limitations that are mentioned on the live sql limitation page. But still you will be able to do all your labs.

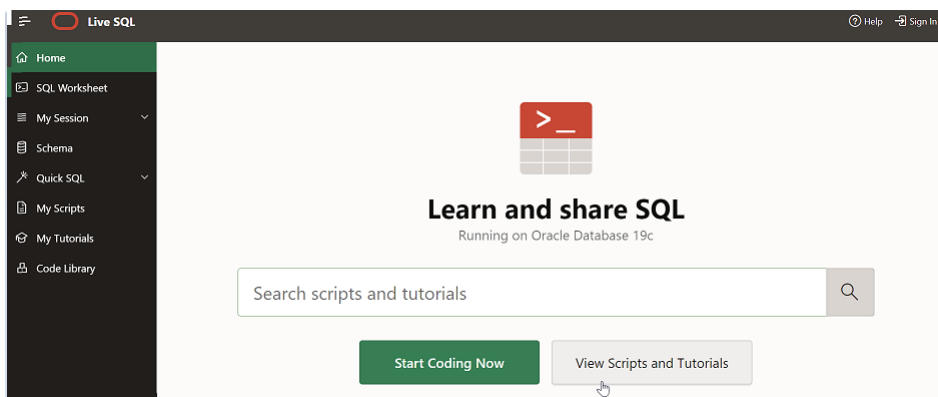


Figure 3.1: Oracle Live SQL

3.2 Oracle Account Creation

To use **oracle live sql** you need to create your **oracle account** on here

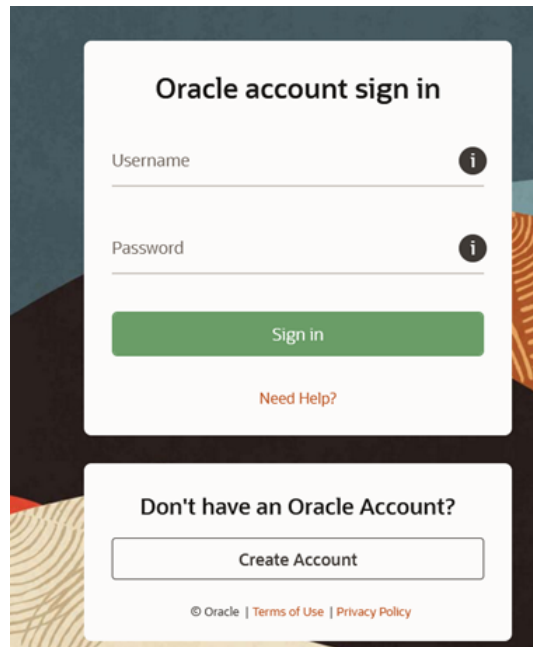
The image shows a web interface for Oracle account management. It features two main sections. The top section, titled "Oracle account sign in", contains input fields for "Username" and "Password", each with an information icon (i) to its right. Below these fields is a green "Sign in" button and a link that says "Need Help?". The bottom section, titled "Don't have an Oracle Account?", contains a "Create Account" button. At the very bottom, there is a footer with the text "© Oracle | Terms of Use | Privacy Policy".

Figure 3.2: Oracle Account

3.3 Uploading labtables.sql script

- Download the labtables.sqlscript from qmplus
- Click on the **Schema** from left menu and then the **UploadScript** button on the top right
- In file give the of labtables.sql and in name simply write labtables or anything else
- Click the **UploadScript** button

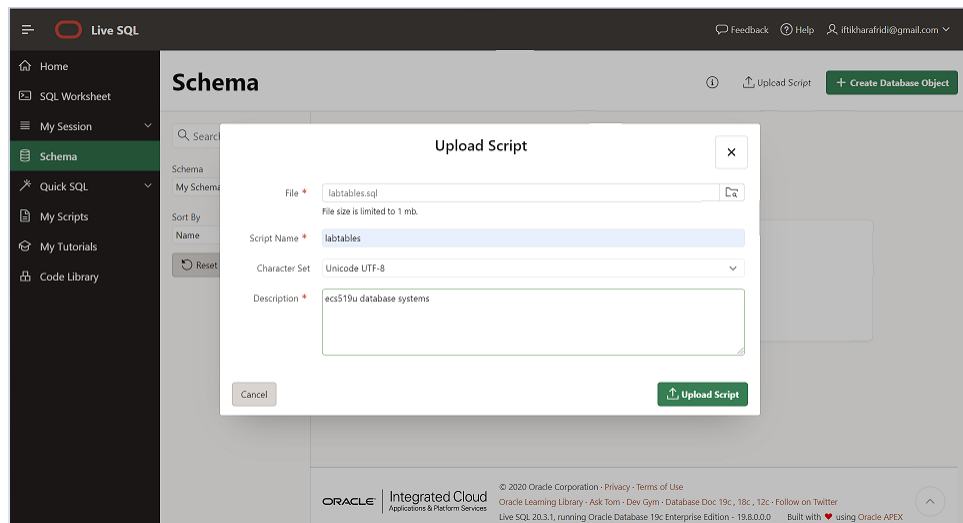


Figure 3.3: labtable script

This will show the script details

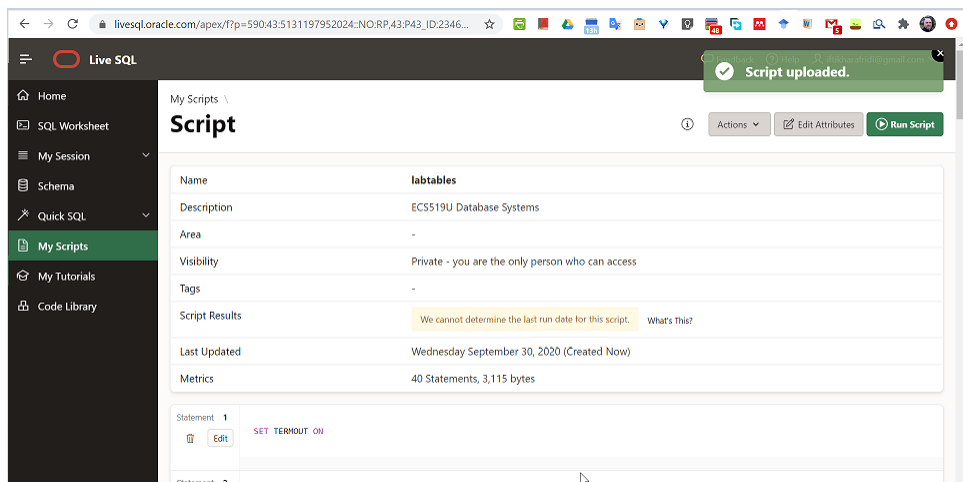


Figure 3.4: script uploaded

After script upload, click the top right **RunScript** button

You should see a couple of messages and warnings. simply ignore them. Last message confirms tables have been created

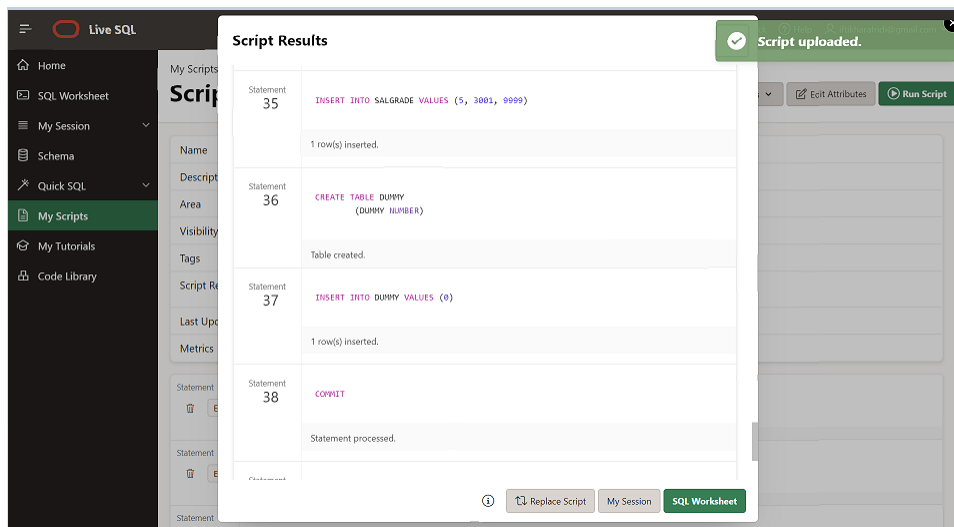


Figure 3.5: Script Run Messages

3.4 Schema/Tables

This creates five tables in your account, called **emp**, **dept**, **bonus**, **salgrade** and **dummy**. Click on **Schema** from left menu and you will be able to see all created tables.

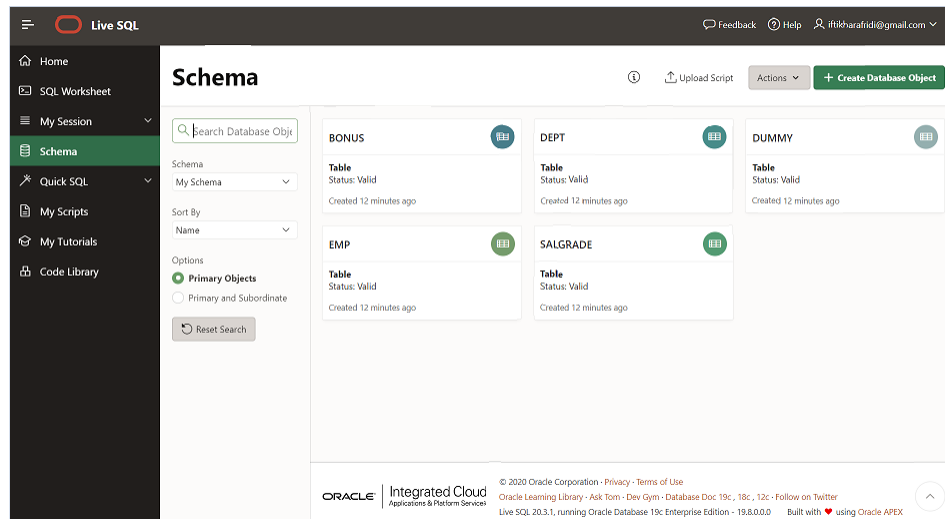


Figure 3.6: Schema/tables

Click on any table and it will give you the table structure.

Schema \ **DEPT**

Syntax Help Actions View All Objects

Show All Table Attributes Columns Indexes Triggers Constraints

Table Attributes

Table Name	DEPT
Status	VALID
Temporary	No
Nested	No
Owner	SQL_GCNMVRIEKHIFVPLQAKJFYVBPF

Columns

#	Column	Type	Length	Precision	Scale	Nullable	Semantics	Comment
1	DEPTNO	NUMBER	22	2	0	Yes		
2	DNAME	VARCHAR2	14			Yes	Byte	
3	LOC	VARCHAR2	13			Yes	Byte	

Indexes

Figure 3.7: Dept table

3.5 Lab Setup Ready

Now you are ready to do the lab exercises

3.6 Basic SQL Help

Meanwhile, a good basic introduction to the language which you can use to assist with the lab exercises can be found here. You can also refer to the lecture notes for this week lecture. You are now ready to try the exercises listed in the SQL Exercise 1 (simple queries file)

Chapter 4

Installing Oracle Database XE on Windows 10

4.1 Download Oracle Database Xe for Windows 10

- Visit Oracle to download the **Oracle Database XE** express edition for Microsoft Windows
- Extract the downloaded **zip** file to a temporary location

4.2 Installation

- Double click on **setup.exe**
- On Welcome to the **InstallShield Wizard** for Oracle Database 18c Express Edition window, click **Next**

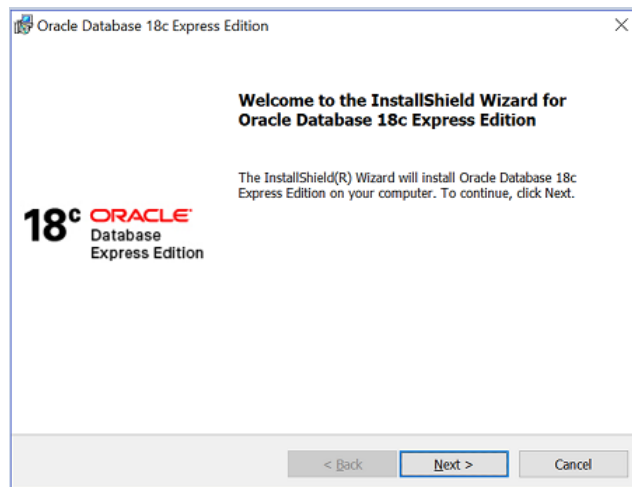


Figure 4.1: InstallShield Wizard for Oracle Database 18c Xe

- In the **License Agreement** window, select I accept the terms in the license agreement and then click
- In the **Choose Destination Location window**, either accept the default or click Change to select a different installation directory. (*Do not select a directory that has spaces in its name.*) Then click **Next**

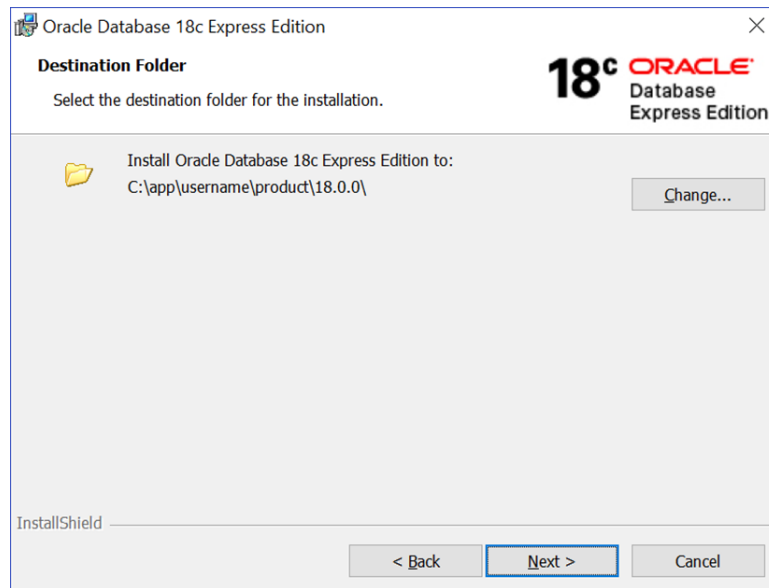


Figure 4.2: Choose Destination Location

- In the **Specify Database Passwords** window, enter and confirm the single database password to use for the **SYS**, **SYSTEM**, and **PDBADMIN** database accounts. Then click **Next**

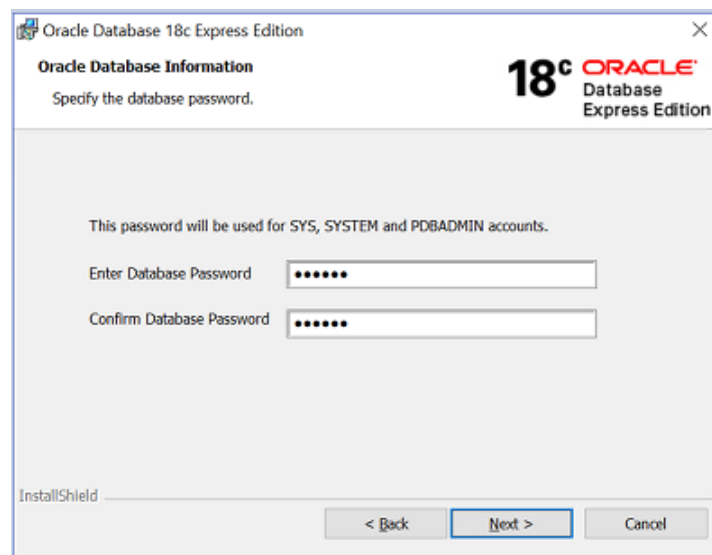


Figure 4.3: Specify Database Passwords

- In the **Summary** window, review the installation settings, and if you are satisfied, click **Install**. Otherwise,

click Back and modify the settings as necessary

– If **Windows Defender Firewall** is activated, then you will be asked to allow network access to the **Java Platform SE binary**. Check both the checkboxes and click the **Allow access** button

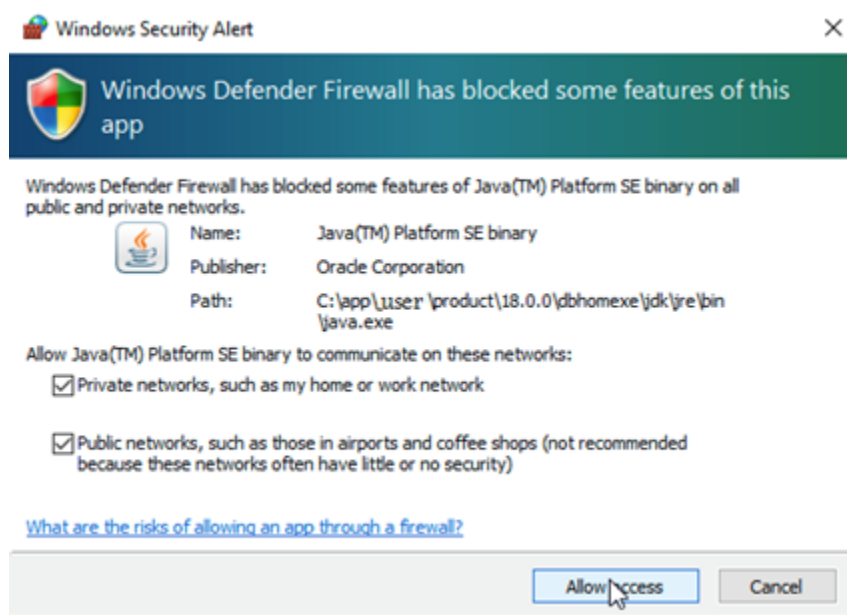


Figure 4.4: Windows Defender Firewall

- When the installation is complete, the **Oracle Database Installed Successfully** window is displayed. Make a note of the connection strings provided for **multitenant container database** and the **pluggable database**, as well as the **EM Express URL**. Click **Finish** to close the installer.

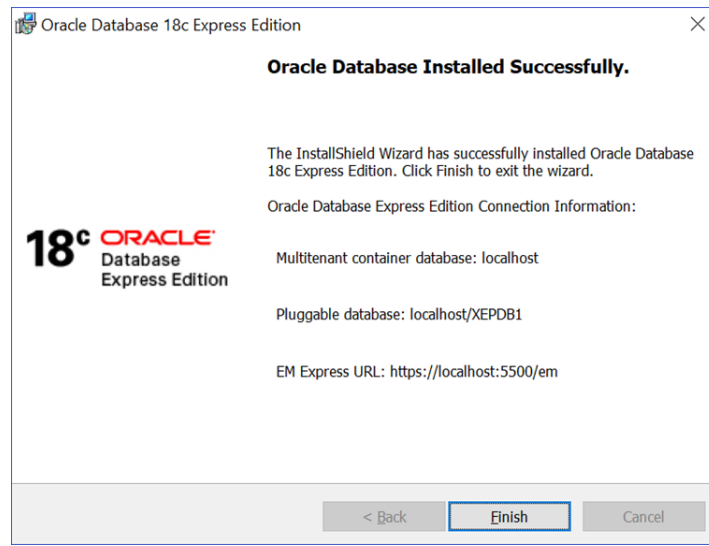


Figure 4.5: Oracle Database Installed Successfully

4.3 SQL*PLUS Command Line Tool

Open SQL*PLUS Command from command prompt or Start Application menu

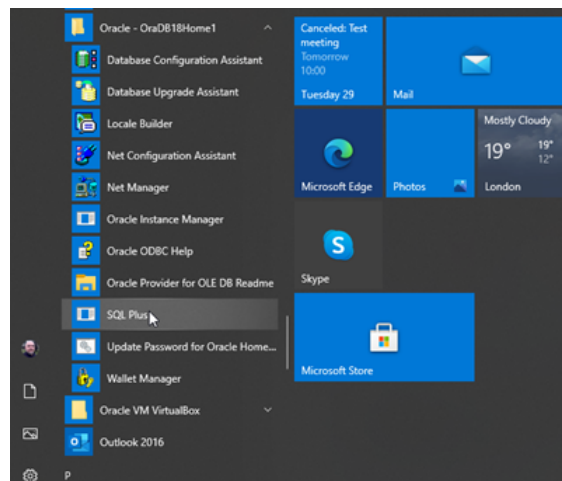


Figure 4.6: Start Menu

4.3.1 Connecting to Oracle using Sys account

Connect to Oracle using **system** user as **sysdba** with the password

```
1 sql> sys as sysdba;
```

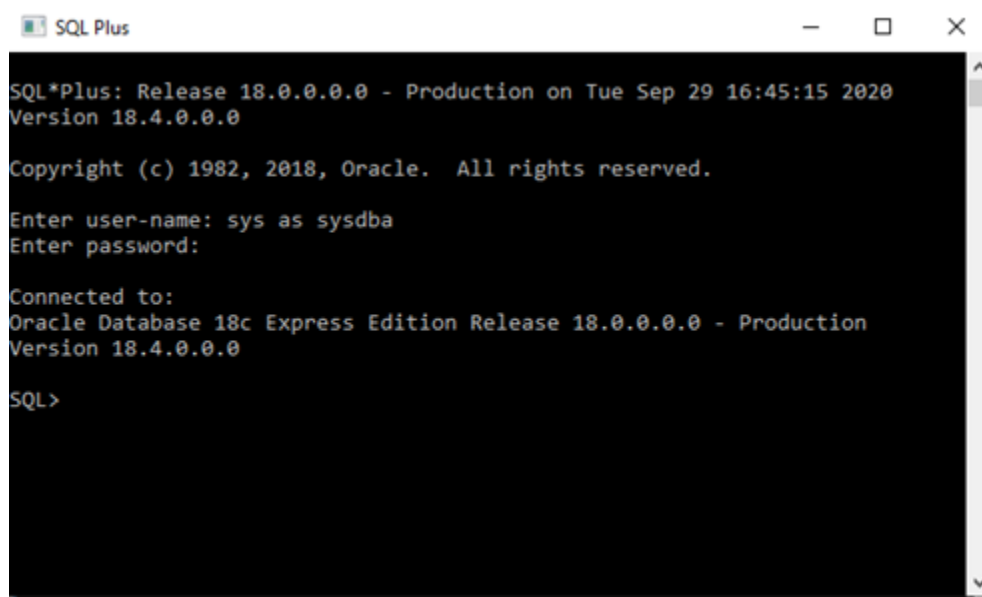


Figure 4.7: Start Menu

4.3.2 Create a local oracle User

Now create a local user/schema

```
1 sql> create user YOUR_USER_NAME identified by YOUR_USER_PASSWORD;  
2  
3 e.g:  
4  
5 sql> create user ecs740p identified by oracle;
```

On first try oracle might give you an **error**

```
1 {"ERROR_at_line_1:  
2 ORA-65096: invalid common user or role name"}
```

Simply apply the following commands and try again

```
1 sql> alter session set "_ORACLE_SCRIPT"=true;  
2 and then  
3 SQL> commit;
```

4.3.3 Grant Permission to User

Now grant permission to the user

```
1 sql> GRANT ALL PRIVILEGES TO ecs740p;
```

4.3.4 Connecting to oracle

Now connect to oracle using created user **ecs740p** and password **oracle**

```
1 sql> connect ecs740p;
```

4.3.5 labtables.sql script

- Download the **labtables.sql** script file from **QMPLUS** and place It in some folder
- Run the script

```
1 sql> start \file_path\labtables.sql;  
2  
3 e.g.  
4  
5 sql> start C:\Users\PC\Downloads\labtables.sql;
```

4.3.6 Schema/Tables

This creates five tables in your account, called **emp**, **dept**, **bonus**, **salgrade** and **dummy**. Use SQLPLUS **Describe** command to check details of a table
For each of the tables, from the SQL prompt, type:

```
1 sql> describe "tablename"
```

4.4 Lab Setup Ready

Now your lab setup is ready. For SQL*PLUS Tool help and commands, please see the SQL*Plus files on QMPLUS

4.5 Basic SQL Help

Meanwhile, a good basic introduction to the language which you can use to assist with the lab exercises can be found here. You can also refer to the lecture notes for this week lecture. You are now ready to try the exercises listed in the SQL Exercise 1 (simple queries file)

Chapter 5

Lab Exercises

5.1 Schema/Tables

The following five tables are created.

EMP: (EMPNO, ENAME, JOB, MGR, HIREDATE, SAL, COMM, DEPTNO);
DEPT: (DEPTNO, DNAME, LOC);
BONUS: (ENAME, JOB, SAL, COMM);
SALGRADE: (GRADE, LOSAL, HISAL);
DUMMY: (DUMMY);

5.2 Exercise-1: SQL

1. List all employees whose salary is between 1,000 and 2,000.
Show employee name, department and salary
2. Display all the different types of occupations
3. List details of employees in departments 10 and 30
4. Display all employees who recruited during 1982, giving their name, department and hiredate
5. List the employees whose names have 'th' or 'll' in them
6. List the department numbers and names in department name order
7. Find the names, jobs, salaries and commissions of all employees who do not have managers
8. List all salesmen in descending order of their commission divided by their salary
9. Calculate the total annual compensation of all salesmen based upon their monthly salary and monthly commission
 $(SAL + COMM) * 12$
10. Find all the salesmen in department 30 who have a salary greater than or equal to \$1,500

5.3 Exercise-2: FUNCTIONS

1. Find out how many managers there are without listing them.
3.
2. Compute the *AVERAGEANNUALSALARY + COMMISSION* for all salesman
 $AVG(SAL + COMM) * 12 = 23,400$
3. Find the highest and lowest paid employees and the difference between them
 $MAX(SAL) = 5,000$
 $MIN(SAL) = 800$
 $MAX(SAL) - MIN(SAL) = 4200$
4. Find the number of characters in the longest department name
 $MAX(LENGTH(DNAME)) = 10$
5. Count the number of people in department 30 who receive a salary and the number of people who receive a commission
 $COUNT(SAL) = 6$
 $COUNT(COMM) = 4$
6. List the average commission of employees who receive a commission and the average commission of all employees (treating employees who do not receive a commission as receiving a zero commission)
 $AVG(COMM) = 550$
 $AVG(NVL(COMM, 0)) = 366.666667$
7. List the average salary of employees that receive a salary, the average commission of employees that receive a commission, the average salary plus commission of only those employees that receive a commission and average salary plus commission of all employees including those who do not receive a commission
 $AVG(SAL) = 1,566.67$
 $AVG(COMM) = 550.00$
 $AVG(SAL + COMM) = 1,950.00$
 $AVG(SAL + NVL(COMM, 0)) = 1,933.22$
8. Compute the daily and hourly salaries for employees in department 30. round the results to the nearest penny. Assume there are 22 working days in a month and 8 working hours in a day
9. Issue the same query as the previous one except that this time truncate (trunc) to the nearest penny rather than round.

5.4 Exercise-3: Dates

1. Select the name, job and hiredate of the employees in dept 20 (format the hiredate column using a picture of mm/dd/yy)
2. Use a picture to format hiredate as: day (day of the week), month (name of the month), dd (day of the month) and yyyy (year).
e.g. wednesday january the 12th, 1983
3. Use the format mask suffix 'th' in a picture (oracle will put a suffix after the field (ddth).)
4. Select the name and hiredate of department 20's employees in the following format:
ename, hiredate
adams, january 12th, 1983, 11 : 05 am

5.5 Exercise-4: JOINS

1. Find the name and salary of employees in dallas
2. Join the dept table to the emp table and show in department number order
3. List all departments that have employees plus those departments that do not have employees
4. List all departments that do not have any employees
5. Find all the employees that earn more than jones using temporary labels to abbreviate table names
6. For each employee whose salary exceeds his manager's salary
List the employee's name and a salary and the manager's name and salary

5.6 Exercise-5: GROUP BY

1. List the department number and average salary of each department
2. Divide all employees into groups by department and by job within department. count the employees in each group and compute each groups average annual salary
3. Issue the same query as above except list the department name rather than the department number
4. List the average annual salary for all job groups having more than 2 employees in the group
5. List all the departments that have at least two clerks
6. Find all departments with an average commission greater than 25% of average salary
7. Find each departments average annual salary for all its employees except the manager's and the president's
8. List the average annual salary for all job groups having more than two employees in a group

5.7 Exercise-6: SUBQUERIES

1. List the name and job of employees who have the same job as jones
2. Find all the employees in department 10 who have a job that is the same as anyone in department 30
3. List the name, job title and salary of employees who have the same job and salary as ford
4. List the name, job and department of employees who have the same job as jones or a salary greater than or equal to ford
5. Find all the employees in department 10 that have a job that is the same as anyone in the sales department
6. Find the employees located in chicago who have the same job as allen. return the results in alphabetical order by employee name
7. Find all employees that earn more than the average salary of employees in their department

5.8 Exercise-7: DATA MANIPULATION

1. Create a new table called loans with columns named lno number(3), empno number(4), type char(1), amnt number(8,2)
2. Insert the following data:

1	LNO , EMPNO , TYPE , AMNT
2	23 , 7499 , M , 20000.00
3	42 , 7499 , C , 2000.00
4	65 , 7844 , M , 3564.20

3. Check you have created 3 new records in loans

4. The loans table must be altered to include another column:
outst number(8,2)
5. Add 10% interest to all m-type loans
6. Remove all loans less than 3,000 pounds
7. Change the name of the loans table to accounts
8. Change the name of column lno to loanno
9. Create a view for use by personnel in department 30 showing
employee name, number, job and hiredate.
10. Use the view to show employees in department 30 having jobs which are not salesman
11. Drop column type from loans table.

5.9 Exercise-8: Advanced

1. Find details of the second most highly salaried employee.
2. Find details of the fifth most highly salaried employee
3. Find all departments which do not employ clerks.
(There are at least three distinct methods for performing this - show at least two)
4. Find the total number of employees in each department.
5. Use a query to display the information required by the following:
The management would like to see details of employees and their salaries after the following measures
All employees earning more than \$1000 will receive an immediate pay cut of \$500, unless this brings their salary below \$1000. If it does, their new salary will be \$1000.
N.B. This is NOT an update - it's a WHAT IF.
Output should have the following format

1	EMPNO	ENAME	AMOUNT DEDUCTED	NEW SALARY
2	-----	-----	-----	-----

6. 6. Construct the following tables in your account

1	PROJNO	DEPTNO
2	-----	-----
3	P1	10
4	P2	30
5	P3	
6		
7	ENAME	DEPTNO
8	-----	-----
9	SMITH	10
10		

11	JONES	20
12	BLOGGS	
13	MILLER	
14		
15	DNAME	DEPTNO
16	-----	-----
17	YORK	10
18	ELY	20
19	RYE	30
20	WELLS	40
21	PERTH	

7. Write two separate queries which will join data from all three tables in an appropriate manner using the foreign keys provided. Your output should consist of eight rows of data if you are to display all the available information. The two queries should show the use of

(i) OUTER JOINS & UNIONS

(ii) UNIONS alone

- Find all employees who joined the company at least six months before their managers
- List lowest paid employees working for each manager (MGR). Exclude any data where the salary is less than 1000. Sort the output on salary.
- Find details of the employee with the lowest salary figure in each department.
- In which year did most people join the company? Display the year and number of employees.
- Create a view showing employees' salaries and jobs which may only be used between the hours of 11:00 and 13:00 each day. Use the view to show details of employees earning more than \$2000.
- Devise a query to show employees with their jobs, but where there are clerks the 'JOB' should be classified as 'INDIAN' and where there are managers the 'JOB' should be shown as 'CHIEF'. Anyone else should be regarded as 'OTHER'.
- Write a query which shows employees with their hiredates and first paydays. The rules about first salary payments are that if the date of hiring is in the first 15 days of the month, the first salary will be paid on the last friday of the same month. If the employee is hired after the first fifteen days, then payment is made on the last friday of the following month.

Your output should have the following appearance.

1	ENAME	HIREDATE	PAYDAY
2	-----	-----	-----
3	SMITH	17-DEC-80	30-JAN-81
4	ALLEN	20-FEB-81	27-MAR-81
5	JONES	02-APR-81	24-APR-81
6	BLAKE	01-MAY-81	29-MAY-81
7	CLARK	09-JUN-81	26-JUN-81
8	TURNER	08-SEP-81	25-SEP-81

9	MARTIN	28-SEP-81	30-OCT-81
10	KING	17-NOV-81	25-DEC-81
11	FORD	03-DEC-81	25-DEC-81
12	SCOTT	09-DEC-82	31-DEC-82
13	ADAMS	12-JAN-83	28-JAN-83

15. From the details available in the following PROJECTS table show names of projects along with the nearest date to today's date on which they have been, are, or will be active.

1	PROJNO	PNAME	START_DATE	END_DATE
2	-----	-----	-----	-----
3	1	ACCOUNTING	01-FEB-90	23-APR-90
4	2	SALESORDERS	31-MAY-90	30-APR-91
5	3	DELIVERIES	02-MAR-91	31-MAR-91
6	4	SUPPORT	20-JUL-88	20-AUG-89
7	5	MIS	15-JAN-90	17-OCT-90
8				
9		(You will need to set up a PROJECTS table containing		
10		this data in appropriate format)		

16. Find the department with the highest total labour costs.
17. Write a query to calculate the length of time in completed months and years that any employee of the users'choice has been with the company.

```

1      (Hint: - you will need to use FLOOR, MOD, MONTHS_BETWEEN
2 and '&' in order to produce the following output.
3
4      SQL> Enter value for employee_name: KING
5
6      ENAME      LENGTH OF SERVICE
7      -----
8      KING       8 YEARS 6 MONTHS

```

Appendix A

Labtables.sql script

Copy and paste the sql queries in some notepad and save-as labtables.sql. or download them from qmplus

The first one is for MySQL.

The second one is a trim version for oracle live sql, because it doesn't support sql*plus commands.

The third one is full version and can be uploaded to on-premises oracle or where sql*plus command are supported.

A.1 Labtables.sql for MySQL

```
1 CREATE DATABASE IF NOT EXISTS 'ecs740p';
2 USE 'ecs740p';
3 select 'Building demonstration tables. Please wait.' as '';
4
5 DROP TABLE IF EXISTS EMP;
6 DROP TABLE IF EXISTS DEPT;
7 DROP TABLE IF EXISTS BONUS;
8 DROP TABLE IF EXISTS SALGRADE;
9 DROP TABLE IF EXISTS DUMMY;
10
11
12 CREATE TABLE EMP
13 (EMPNO SMALLINT NOT NULL,
14  ENAME VARCHAR(10),
15  JOB VARCHAR(9),
16  MGR SMALLINT,
17  HIREDATE DATETIME,
18  SAL DECIMAL(7, 2),
19  COMM DECIMAL(7, 2),
20  DEPTNO TINYINT);
21
22 INSERT INTO EMP VALUES
23 (7369, 'SMITH', 'CLERK', 7902,
24  STR_TO_DATE('17-DEC-1980', '%d-%M-%Y'), 800, NULL, 20);
25 INSERT INTO EMP VALUES
26 (7499, 'ALLEN', 'SALESMAN', 7698,
27  STR_TO_DATE('20-FEB-1981', '%d-%M-%Y'), 1600, 300, 30);
28 INSERT INTO EMP VALUES
29 (7521, 'WARD', 'SALESMAN', 7698,
30  STR_TO_DATE('22-FEB-1981', '%d-%M-%Y'), 1250, 500, 30);
31 INSERT INTO EMP VALUES
32 (7566, 'JONES', 'MANAGER', 7839,
33  STR_TO_DATE('2-APR-1981', '%d-%M-%Y'), 2975, NULL, 20);
34 INSERT INTO EMP VALUES
35 (7654, 'MARTIN', 'SALESMAN', 7698,
36  STR_TO_DATE('28-SEP-1981', '%d-%M-%Y'), 1250, 1400, 30);
37 INSERT INTO EMP VALUES
38 (7698, 'BLAKE', 'MANAGER', 7839,
39  STR_TO_DATE('1-MAY-1981', '%d-%M-%Y'), 2850, NULL, 30);
40 INSERT INTO EMP VALUES
41 (7782, 'CLARK', 'MANAGER', 7839,
42  STR_TO_DATE('9-JUN-1981', '%d-%M-%Y'), 2450, NULL, 10);
43 INSERT INTO EMP VALUES
44 (7788, 'SCOTT', 'ANALYST', 7566,
45  STR_TO_DATE('09-DEC-1982', '%d-%M-%Y'), 3000, NULL, 20);
46 INSERT INTO EMP VALUES
47 (7839, 'KING', 'PRESIDENT', NULL,
48  STR_TO_DATE('17-NOV-1981', '%d-%M-%Y'), 5000, NULL, 10);
49 INSERT INTO EMP VALUES
50 (7844, 'TURNER', 'SALESMAN', 7698,
51  STR_TO_DATE('8-SEP-1981', '%d-%M-%Y'), 1500, 0, 30);
52 INSERT INTO EMP VALUES
53 (7876, 'ADAMS', 'CLERK', 7788,
54  STR_TO_DATE('12-JAN-1983', '%d-%M-%Y'), 1100, NULL, 20);
55 INSERT INTO EMP VALUES
56 (7900, 'JAMES', 'CLERK', 7698,
57  STR_TO_DATE('3-DEC-1981', '%d-%M-%Y'), 950, NULL, 30);
58 INSERT INTO EMP VALUES
59 (7902, 'FORD', 'ANALYST', 7566,
60  STR_TO_DATE('3-DEC-1981', '%d-%M-%Y'), 3000, NULL, 20);
61 INSERT INTO EMP VALUES
```


A.2 Labtables.sql for Oracle Live SQL

```
1 CREATE TABLE EMP
2   (EMPNO NUMBER(4) NOT NULL,
3     ENAME VARCHAR2(10),
4     JOB VARCHAR2(9),
5     MGR NUMBER(4),
6     HIREDATE DATE,
7     SAL NUMBER(7, 2),
8     COMM NUMBER(7, 2),
9     DEPTNO NUMBER(2));
10
11 INSERT INTO EMP VALUES
12   (7369, 'SMITH', 'CLERK', 7902,
13     TO_DATE('17-DEC-1980', 'DD-MON-YYYY'), 800, NULL, 20);
14 INSERT INTO EMP VALUES
15   (7499, 'ALLEN', 'SALESMAN', 7698,
16     TO_DATE('20-FEB-1981', 'DD-MON-YYYY'), 1600, 300, 30);
17 INSERT INTO EMP VALUES
18   (7521, 'WARD', 'SALESMAN', 7698,
19     TO_DATE('22-FEB-1981', 'DD-MON-YYYY'), 1250, 500, 30);
20 INSERT INTO EMP VALUES
21   (7566, 'JONES', 'MANAGER', 7839,
22     TO_DATE('2-APR-1981', 'DD-MON-YYYY'), 2975, NULL, 20);
23 INSERT INTO EMP VALUES
24   (7654, 'MARTIN', 'SALESMAN', 7698,
25     TO_DATE('28-SEP-1981', 'DD-MON-YYYY'), 1250, 1400, 30);
26 INSERT INTO EMP VALUES
27   (7698, 'BLAKE', 'MANAGER', 7839,
28     TO_DATE('1-MAY-1981', 'DD-MON-YYYY'), 2850, NULL, 30);
29 INSERT INTO EMP VALUES
30   (7782, 'CLARK', 'MANAGER', 7839,
31     TO_DATE('9-JUN-1981', 'DD-MON-YYYY'), 2450, NULL, 10);
32 INSERT INTO EMP VALUES
33   (7788, 'SCOTT', 'ANALYST', 7566,
34     TO_DATE('09-DEC-1982', 'DD-MON-YYYY'), 3000, NULL, 20);
35 INSERT INTO EMP VALUES
36   (7839, 'KING', 'PRESIDENT', NULL,
37     TO_DATE('17-NOV-1981', 'DD-MON-YYYY'), 5000, NULL, 10);
38 INSERT INTO EMP VALUES
39   (7844, 'TURNER', 'SALESMAN', 7698,
40     TO_DATE('8-SEP-1981', 'DD-MON-YYYY'), 1500, 0, 30);
41 INSERT INTO EMP VALUES
42   (7876, 'ADAMS', 'CLERK', 7788,
43     TO_DATE('12-JAN-1983', 'DD-MON-YYYY'), 1100, NULL, 20);
44 INSERT INTO EMP VALUES
45   (7900, 'JAMES', 'CLERK', 7698,
46     TO_DATE('3-DEC-1981', 'DD-MON-YYYY'), 950, NULL, 30);
47 INSERT INTO EMP VALUES
48   (7902, 'FORD', 'ANALYST', 7566,
49     TO_DATE('3-DEC-1981', 'DD-MON-YYYY'), 3000, NULL, 20);
50 INSERT INTO EMP VALUES
51   (7934, 'MILLER', 'CLERK', 7782,42
52     TO_DATE('23-JAN-1982', 'DD-MON-YYYY'), 1300, NULL, 10);
53
54 CREATE TABLE DEPT
55   (DEPTNO NUMBER(2),
56     DNAME VARCHAR2(14),
57     LOC VARCHAR2(13) );
58
59 INSERT INTO DEPT VALUES (10, 'ACCOUNTING', 'NEW_YORK');
60 INSERT INTO DEPT VALUES (20, 'RESEARCH', 'DALLAS');
61 INSERT INTO DEPT VALUES (30, 'SALES', 'CHICAGO');
```


A.3 Labtables.sql script for sql*plus

```
1
2 SET TERMOUT ON
3 PROMPT Building demonstration tables. Please wait.
4 SET TERMOUT OFF
5
6 DROP TABLE EMP;
7 DROP TABLE DEPT;
8 DROP TABLE BONUS;
9 DROP TABLE SALGRADE;
10 DROP TABLE DUMMY;
11
12 CREATE TABLE EMP
13     (EMPNO NUMBER(4) NOT NULL,
14      ENAME VARCHAR2(10),
15      JOB VARCHAR2(9),
16      MGR NUMBER(4),
17      HIREDATE DATE,
18      SAL NUMBER(7, 2),
19      COMM NUMBER(7, 2),
20      DEPTNO NUMBER(2));
21
22 INSERT INTO EMP VALUES
23     (7369, 'SMITH', 'CLERK', 7902,
24      TO_DATE('17-DEC-1980', 'DD-MON-YYYY'), 800, NULL, 20);
25 INSERT INTO EMP VALUES
26     (7499, 'ALLEN', 'SALESMAN', 7698,
27      TO_DATE('20-FEB-1981', 'DD-MON-YYYY'), 1600, 300, 30);
28 INSERT INTO EMP VALUES
29     (7521, 'WARD', 'SALESMAN', 7698,
30      TO_DATE('22-FEB-1981', 'DD-MON-YYYY'), 1250, 500, 30);
31 INSERT INTO EMP VALUES
32     (7566, 'JONES', 'MANAGER', 7839,
33      TO_DATE('2-APR-1981', 'DD-MON-YYYY'), 2975, NULL, 20);
34 INSERT INTO EMP VALUES
35     (7654, 'MARTIN', 'SALESMAN', 7698,
36      TO_DATE('28-SEP-1981', 'DD-MON-YYYY'), 1250, 1400, 30);
37 INSERT INTO EMP VALUES
38     (7698, 'BLAKE', 'MANAGER', 7839,
39      TO_DATE('1-MAY-1981', 'DD-MON-YYYY'), 2850, NULL, 30);
40 INSERT INTO EMP VALUES
41     (7782, 'CLARK', 'MANAGER', 7839,
42      TO_DATE('9-JUN-1981', 'DD-MON-YYYY'), 2450, NULL, 10);
43 INSERT INTO EMP VALUES
44     (7788, 'SCOTT', 'ANALYST', 7566,
45      TO_DATE('09-DEC-1982', 'DD-MON-YYYY'), 3000, NULL, 20);
46 INSERT INTO EMP VALUES
47     (7839, 'KING', 'PRESIDENT', NULL,
48      TO_DATE('17-NOV-1981', 'DD-MON-YYYY'), 5000, NULL, 10);
49 INSERT INTO EMP VALUES
50     (7844, 'TURNER', 'SALESMAN', 7698,
51      TO_DATE('8-SEP-1981', 'DD-MON-YYYY'), 1500, 0, 30);
52 INSERT INTO EMP VALUES
53     (7876, 'ADAMS', 'CLERK', 7788,
54      TO_DATE('12-JAN-1983', 'DD-MON-YYYY'), 1100, NULL, 20);
55 INSERT INTO EMP VALUES
56     (7900, 'JAMES', 'CLERK', 7698,
57      TO_DATE('3-DEC-1981', 'DD-MON-YYYY'), 950, NULL, 30);
58 INSERT INTO EMP VALUES
59     (7902, 'FORD', 'ANALYST', 7566,
60      TO_DATE('3-DEC-1981', 'DD-MON-YYYY'), 3000, NULL, 20);
61 INSERT INTO EMP VALUES
```

Appendix B

Oracle Database on Mac OS using Docker



Figure B.1: Oracle, Mac OS and Docker

B.1 No Oracle for Mac OS

Oracle database is only available for Windows and Linux. There is no installer file for Mac, but can be run on Mac using virtual environment. Here we will use the **Docker Container** to run oracle on mac os,

B.2 Git Command

- Visit [Git-SCM](#) to download the [GIT](#) for Mac OS
- Click the [Binary Installer](#) link

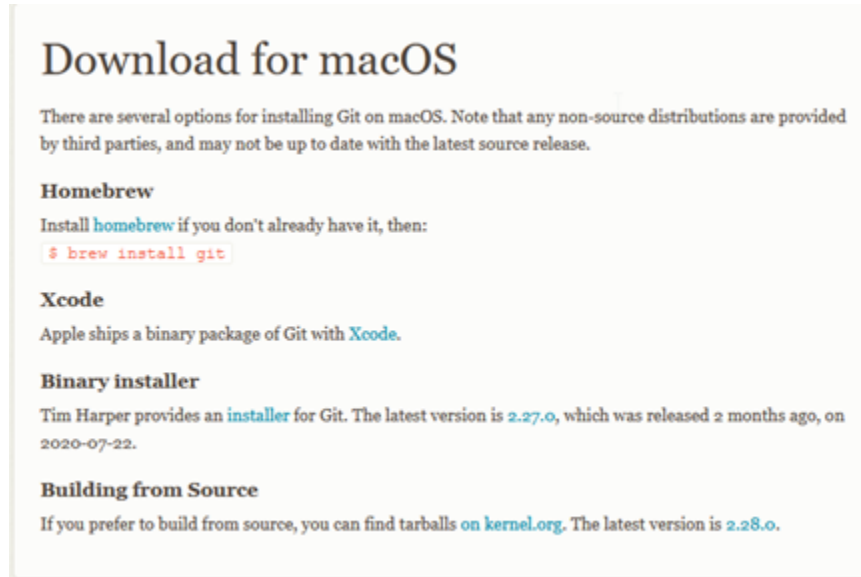


Figure B.2: Git-SCM

B.3 Docker for Mac

- Visit Docker and create an account
- Download Docker-Desktop for Mac OS
- Double click the downloaded **Docker.dmg** file
- Drag and drop the Docker icon into the Application folder

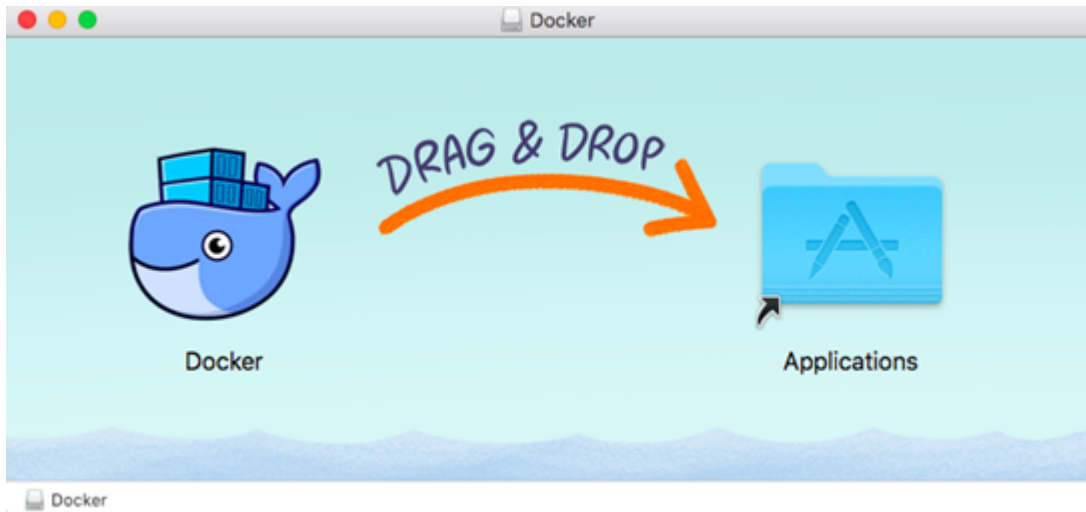


Figure B.3: Docker-Desktop Installation

- Run Docker from Application folder
- Docker icon will appear on top right of screen
- You can check if it's running by clicking on it

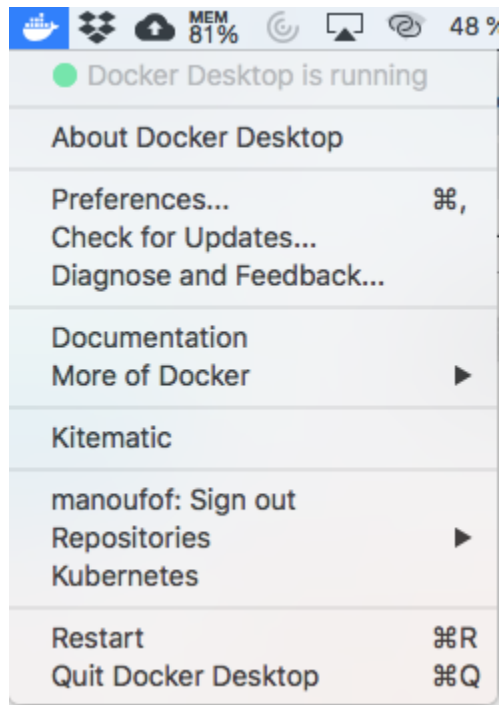
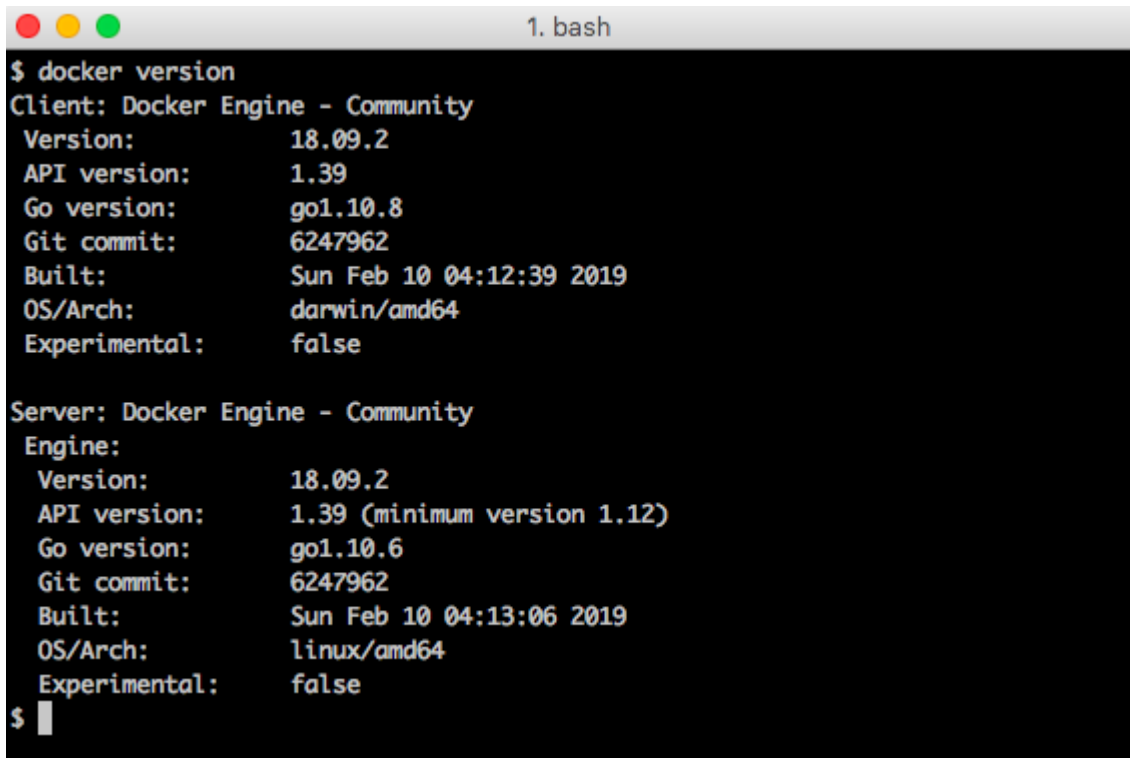


Figure B.4: Docker Options

B.4 Terminal Window

Open **Terminal** window and test the docker version command

```
1 $ docker version
```



```
1. bash
$ docker version
Client: Docker Engine - Community
 Version:           18.09.2
 API version:       1.39
 Go version:        go1.10.8
 Git commit:        6247962
 Built:             Sun Feb 10 04:12:39 2019
 OS/Arch:           darwin/amd64
 Experimental:      false

Server: Docker Engine - Community
 Engine:
  Version:           18.09.2
  API version:       1.39 (minimum version 1.12)
  Go version:        go1.10.6
  Git commit:        6247962
  Built:             Sun Feb 10 04:13:06 2019
  OS/Arch:           linux/amd64
  Experimental:      false
$
```

Figure B.5: Terminal Window

B.5 Pulling Oracle Database Image

Now pull the oracle database image from **Docker-HUB**

```
1 $ docker pull truevoly/oracle-12c
```

You will see a message is printed once completed:

```
1 Status: Downloaded newer for truevoly/oracle-12c:latest
```

Note: The image is in Giga Bytes, so it takes time to download, depending on your internet connection

B.5.1 Folder for Database

Create a folder for the oracle data

```
1 $ mkdir ~/oracle_data
```

B.5.2 Oracle Container

Create a container of oracle database

```
1 $ docker run -d --name oracle -p 8080:8080 -p 1521:1521
2 -v ~/oracle_data:/u01/app/oracle truevoly/oracle-12c
```

A hash will be generated at the end of the execution.

B.5.3 Status check using docker ps command

Check the status of the oracle container

```
1 $ docker ps -a
```

B.5.4 Database Initialisation and log command

It takes a lot of time to initialise the database. You can monitor the status/progress using docker log command.

```
1 $ docker logs -f oracle
```

Here **oracle** is the name of the container, it can be different. You can use the **Container ID** as well.

B.6 SQL*PLUS Command Line Tool

You can connect to the Oracle using **SQL*Plus** tool.

```
1 $ docker exec -it oracle bash
2 -c "source_/home/oracle/.bashrc;_sqlplus_/nolog"
```

B.6.1 Connecting to Oracle using Sys account

Connect to Oracle using **SYS** user as **sysdba** with the default password **oracle**

```
1 $ connect sys as sysdba;
```

B.6.2 Create a local oracle User

Now create a local **user/schema** and **grant** the necessary permissions.

```
1      $ create user YOUR_USER_NAME identified by YOUR_USER_PASSWORD;  
2  
3      e.g:  
4  
5      $ create user ecs740p identified by oracle;
```

On first try oracle might give you an **error**

```
1      "ERROR at line 1:  
2      ORA-65096: invalid common user or role name".
```

Simply apply the following commands and try again

```
1      sql> alter session set "_ORACLE_SCRIPT"=true;  
2      and then  
3      SQL> commit;
```

B.6.3 Grant Permission to User

Now grant permission to the user

```
1      sql> GRANT ALL PRIVILEGES TO ecs740p;
```

B.6.4 Connecting to oracle

Now connect to oracle using created user **ecs740p** and password **oracle**

```
1      sql> connect ecs740p;
```

B.6.5 labtables.sql script

- Download the **labtables.sql** script file from qmplus and place It in ~/oracle_data folder that is created
- Run the script

```
1      sql> start ./file_path/labtables.sql;  
2  
3      e.g.  
4  
5      sql> start /u01/app/oracle/labtables.sql;
```

B.6.6 Schema/Tables

This creates five tables in your account, called **emp**, **dept**, **bonus**, **salgrade** and **dummy**. Use SQLPLUS **Describe** command to check details of a table
For each of the tables, from the SQL prompt, type:

1

```
sql> describe "tablename"
```

B.7 Lab Setup Ready

Now your lab setup is ready. For SQL*PLUS Tool help and commands, please see the SQL*Plus files on QMPLUS

B.8 Basic SQL Help

Meanwhile, a good basic introduction to the language which you can use to assist with the lab exercises can be found here. You can also refer to the lecture notes for this week lecture. You are now ready to try the exercises listed in the SQL Exercise 1 (simple queries file)